

Extended Evaluation Details and Supplementary Results

S1 Dataset Details

In this section, we present a comprehensive characterization of the datasets employed in our evaluation. The collection consists of ten real-world BGP anomaly events, divided into two groups: eight datasets serve as the benchmark for our comparative detection performance analysis, while the remaining two (categorized as “Other”) are utilized specifically for in-depth case studies (§5.4 of the main paper) and discussions (§6 of the main paper). We used datasets sourced from RouteViews [16] and RIPE RIS [15], and our framework is fully compatible with other sources, such as bgproutes.io [4], to extend visibility.

We collect candidate incidents from incident reports, security analyses, and BGP monitoring platforms. An incident is retained only when its time window, culprit AS, and affected prefixes or origin ASes can be cross-validated from public BGP data and public reports. This protocol is applied before evaluating Parade and is independent of Parade’s detection outcomes. Thus, the dataset is intended as a publicly verifiable and challenge-oriented benchmark.

Table S1 summarizes the detailed specifications of these incidents. Notably, unlike prior studies that often relied on older or exclusively large-scale datasets [9, 17], we prioritized recent real-world events that reflect the current challenges in routing anomaly detection.

Furthermore, recognizing that the scope of an anomaly significantly impacts detection difficulty, we explicitly distinguish between large-scale and small-scale incidents based on the number of affected prefixes and ASes. For instance, the *Cloudflare* incident is annotated as a small-scale route leak, as it primarily affected prefixes within a specific /36 IPv6 block (2a03:2880:f::/36). This categorization allows us to assess the detection capabilities of different methods under varying conditions, ranging from massive route leaks to targeted, low-volume manipulations. The table also reports the culprit AS involved in each event and the final dataset size measured after the deduplication process.

It is noteworthy that the volume of BGP data has been increasing rapidly over time. As of January 2026, the size of the routing data (including both RIBs and updates within a 24-hour period) after deduplication has approached 70 GB, which is more than three times the size of the 2017 data. With the continuous improvement of observation infrastructure [4], it is foreseeable that the scale of BGP data will continue to grow in the future.

S2 Details of the Propagation Fingerprinting

Each AS link $(u, v) \in C_p$ contributes to the fingerprint and is mapped to a feature space via a composite hash function $H : \mathbb{R}^2 \rightarrow \{0, 1\}^{128}$, formed by concatenating independently computed high and low 64-bit segments:

$$H(u, v) = \phi_{\text{high}}(u, v) \parallel \phi_{\text{low}}(u, v).$$

To ensure feature orthogonality between segments, we utilize distinct linear congruential generators and bitwise transformations. Specifically, a bitmask XOR preprocessing is applied to the high bits to disrupt linear correlations with the low bits.

We then maintain an accumulator $\mathbf{x} \in \mathbb{R}^{128}$. For every $(u, v) \in C_p$, $\mathbf{x}$ is updated based on the hash bits, weighted by parameter α and the source node’s depth $h_p(u)$:

$$\mathbf{x}[k] \leftarrow \mathbf{x}[k] + \begin{cases} e^{-\alpha h_p(u)}, & \text{if } H(u, v)[k] = 1 \\ -e^{-\alpha h_p(u)}, & \text{if } H(u, v)[k] = 0 \end{cases}.$$

After traversing all links, we apply a sign function to the accumulator $\mathbf{x}$ to obtain the binary propagation fingerprint $\mathbf{f}_p$:

$$\mathbf{f}_p[k] = \begin{cases} 1, & \text{if } \mathbf{x}[k] > 0 \\ 0, & \text{if } \mathbf{x}[k] \leq 0 \end{cases}.$$

According to Equation 3 of the main paper, the similarity between profiles is transformed rather than directly measured by the Hamming distance. This is because the expected value of Hamming distance between two L -bit random variables is $L/2$. To focus on cases with genuine positive correlation, we shift the expected value to 0 and rescale it to the range $[0, 1]$.

S3 Sampling Strategy Analysis

We provide an empirical analysis of the *Structured Batch Sampling* strategy described in §3.5 of the main paper, utilizing statistical metrics derived from RouteViews RIB snapshots (2025.02–2025.05). The key parameters are defined as follows:

- Global active AS count: $|\mathcal{A}| \approx 75,000$.
- Average related AS count: $\mathbb{E}[|\mathcal{A}_p|] \approx 333.6$.
- Maximum related AS count: $\max(|\mathcal{A}_p|) = 931$.
- Sample sizes: Positive $M = 128$, Negative $3M = 384$.
- Convergence epochs: $T = 600$.

False Negative Rate (FNR). A false negative occurs when a sampled “unrelated” AS from the global set $\mathcal{A}$ actually belongs to the target profile’s related set $\mathcal{A}_p$. Since negative sampling is uniform over $\mathcal{A}$, the single-sample false negative probability is $P_{\text{FN}}(p) = |\mathcal{A}_p|/|\mathcal{A}|$. Substituting the empirical values yields an average FNR of $P_{\text{FN}}^{\text{avg}} \approx 0.44\%$ and a worst-case FNR of $P_{\text{FN}}^{\text{max}} \approx 1.24\%$. These magnitudes confirm that noise from false negatives is statistically negligible relative to the dominant gradient signals provided by the 384 negative samples per batch.

Positive Sample Coverage. We analyze the topological coverage given a fixed sample size $M = 128$. The per-epoch coverage ratio is defined as $C(p) = \min(1, M/|\mathcal{A}_p|)$. This results in an average coverage of $C_{\text{avg}} \approx 38.37\%$ and a worst-case lower bound of $C_{\text{min}} \approx 13.75\%$. Although partial sampling is performed per epoch, the probability of a specific related AS $a \in \mathcal{A}_p$ being omitted decays exponentially over time. Even in the worst-case scenario, the omission probability after $T = 50$ epochs is:

$$P(\text{miss}) = (1 - C_{\text{min}})^T \approx (0.862)^{50} < 6 \times 10^{-4}.$$

Furthermore, the expected selection frequency for any specific AS over the full training course is:

$$\mathbb{E}[\text{count}] = T \cdot C_{\text{min}} = 600 \times 0.1375 = 82.5.$$

Thus, comprehensive topological coverage is guaranteed at an early stage of training.

Table S1: Summary of historical BGP anomaly datasets. The incidents are categorized into origin hijack (OH), route leak (RL), and path manipulation (PM). Datasets marked with an asterisk (*) denote small-scale incidents. The “Other” category includes datasets not used in comparison experiments. Dataset size is measured after deduplication. Aff. denotes Affected.

Category	Dataset Name	Date	Start Time	End Time	Culprit AS	Aff. Pfxs.	Aff. ASes	Size (GB)
OH	Rostelecom [5]	2020-04-01	18:00	21:00	12389	> 5000	> 100	46.22
	Vodafone [1, 3]	2021-04-16	12:00	15:00	55410	> 10000	> 200	53.27
	Celer bridge* [11]	2022-08-17	18:00	22:00	209243	1	1	63.17
RL	Safehost [2]	2019-06-06	08:00	13:00	21217	> 10000	> 100	34.80
	TIC [10]	2023-03-12	12:00	16:00	49666	> 10000	> 100	66.97
	Voxility [12]	2025-04-01	06:00	10:00	3223	> 10000	> 100	68.29
	Cloudflare* [7]	2026-01-22	20:00	23:59	13335	~ 500	1	69.90
PM	Klayswap* [8, 11]	2022-02-03	00:00	05:00	6461 (via 9457)	2	1	59.38
Other	Rostelecom2 [14]	2017-04-26	20:00	23:59	12389	> 100	> 20	22.36
	CANTV [6, 13]	2026-01-02	14:00	18:00	-	0	0	69.88

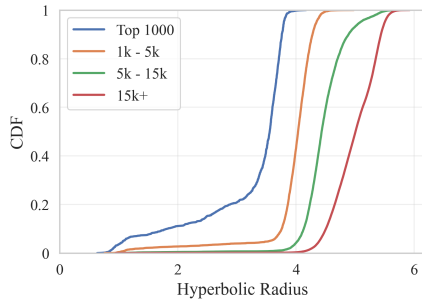


Figure S1: The hyperbolic radius distribution of AS embeddings with different AS ranks.

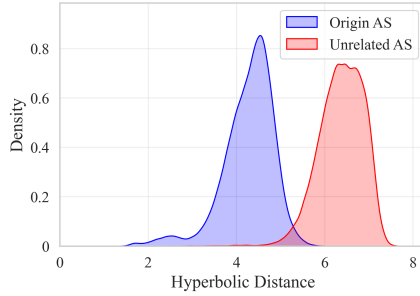


Figure S2: The PDF of hyperbolic distances between the profile embeddings and their origin AS embeddings for positive versus negative pairs.

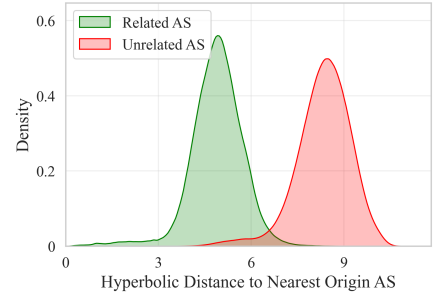


Figure S3: Distribution of hyperbolic distances from origin ASes to propagating ASes in the propagation graphs versus unrelated ASes.

Negative Sample Diversity. The model’s discriminative power relies on diverse contrastive samples. With 384 negative samples per epoch, the total volume of negative contrasts generated for a single profile is $384 \times 600 = 230,400 \gg |\mathcal{A}|$. Consequently, the expected number of times any specific AS serves as a negative sample for a given profile is:

$$\mathbb{E}[\text{visits}] = \frac{3M \cdot T}{|\mathcal{A}|} \approx \frac{230,400}{75,000} \approx 3.07.$$

This ensures that the model contrasts every profile against the full global topology multiple times, preventing the embedding space from degenerating into trivial solutions.

S4 Additional Result Analysis

We further analyze the geometric properties of the learned embedding space, and how the learned semantics and relationships support anomaly detection.

S4.1 Additional Embedding Geometry

As a complement to the structural visualization, Figure S1 further quantifies the hierarchical relationship by plotting AS rank distribution against hyperbolic radius. We observe a general trend where

higher-ranked ASes tend to occupy smaller radii, confirming that the embedding captures structural hierarchy in the routing system. These results support the claim in § 3.6.1 of the main paper that the learned space encodes routing hierarchy in a data-driven manner.

S4.2 Origin and Transit Distance Analysis

Here we analyze the distance distributions that directly inform the design of the anomaly detection in § 4.4 of the main paper.

Figure S2 analyzes the hyperbolic distances from profile embeddings to their origin ASes versus unrelated ASes, while Figure S3 examines distances from origin ASes to related propagating ASes versus unrelated ASes. Although the separations are less absolute than the aggregate distribution in Figure 6 of the main paper, we observe significant distributional shifts: legitimate profile-origin and related AS-origin pairs consistently exhibit smaller distances. This geometric structure directly informs the design of the anomaly detection mechanism that incorporates both profile-origin and origin-AS distance, enabling a more comprehensive capture of the diverse manifestations of anomalous behavior.

Figure S4 shows the cumulative distribution of hyperbolic distances between profiles and ASes at different propagation depths. We observe a consistent trend: ASes appearing further downstream

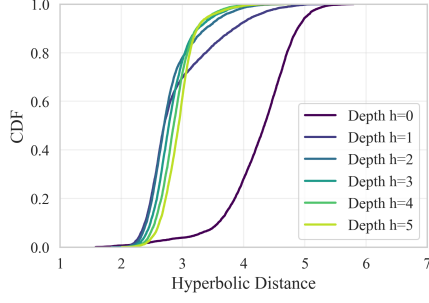


Figure S4: The CDF of hyperbolic distances between profile embeddings and AS embeddings at different propagation depths.

tend to have larger distances to the profile embedding. Importantly, origin ASes are not always the closest entities in the embedding space. Certain transit ASes can exhibit comparable or even stronger proximity when they dominate the propagation structure. This suggests that semantic similarity is governed by propagation behavior rather than origin identity, and provides additional evidence that the embedding space captures propagation-aware semantics.

S4.3 Latent Semantic Generalization

Here, we explore the encoder’s capability to generalize beyond observed routing data to capture latent routing policies. A persistent challenge in BGP anomaly detection is distinguishing “unseen but legal” paths (e.g., backup links) from genuine anomalies. In Figure S5, we quantify the number of ASes falling within the acceptance threshold (defined as the 99.95th percentile of the positive distribution in Figure 6 of the main paper). As the count of observed related ASes increases, the number of potential legitimate ASes rises gradually, stabilizing between several hundred and a thousand. Given the global routing table size (75k+ ASes), this demonstrates that the model effectively generalizes to discover latent backup paths without becoming overly permissive.

S4.4 Detailed Inter-Profile Semantic Analysis

We evaluate the semantic fidelity of the learned prefix embeddings by examining the correlation between pairwise hyperbolic distance and propagation fingerprint similarity.

Figure S7 indicates that hyperbolic distance decreases as fingerprint similarity increases. We observe a distinct *bifurcation* around a similarity threshold of 0.5. This implies that the encoder learns a non-linear boundary for *co-propagation affinity*, rather than performing a simple linear regression.

This separation is quantified in Figure S8, where prefix pairs are categorized into three classes: *siblings* ($s \geq 0.5$), *neighbors* ($0.2 \leq s < 0.5$), and *strangers* ($s < 0.2$). The CDFs demonstrate that *strangers* and *neighbors* share indistinguishable distributions with large hyperbolic distances, suggesting semantic dissociation in the embedding space. In contrast, *siblings* exhibit a significant leftward shift toward proximity. These results confirm that the

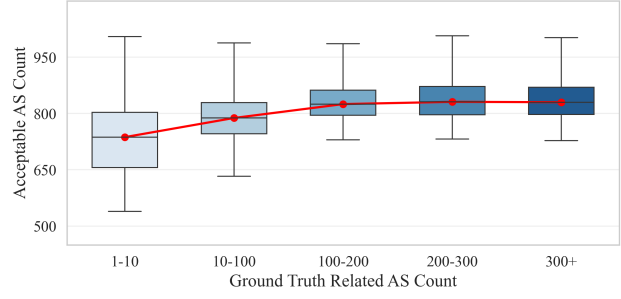


Figure S5: The acceptable AS count within the positive sample distance threshold for each profile, including the latent related ASes. The whiskers represent the 1st-99th percentiles, and the central line indicates the median.

hyperbolic metric captures high-fidelity propagation semantics, effectively distinguishing true algorithmic variants (prefixes sharing core routing policies) from coincidental similarities.

In this section, we provide a quantitative demonstration of how the Poincaré ball model accommodates the exponential growth of nodes (e.g., stub ASes) near the boundary without overcrowding. We compare the Euclidean space required to represent a unit of semantic difference (hyperbolic distance) at different radial positions.

To isolate the relationship between Euclidean distance and hyperbolic distance, let us consider two nodes $\mathbf{u}$ and $\mathbf{v}$ located at the same radial distance r from the origin (i.e., $\|\mathbf{u}\| = \|\mathbf{v}\| = r$). Let $\delta = \|\mathbf{u} - \mathbf{v}\|$ be the Euclidean distance between them. We aim to find the required δ such that their hyperbolic distance is exactly 1 unit (i.e., $d_D(\mathbf{u}, \mathbf{v}) = 1$).

Substituting these conditions into the hyperbolic distance formula, we have:

$$1 = \cosh^{-1} \left(1 + \frac{2\delta^2}{(1-r^2)^2} \right).$$

Applying the hyperbolic cosine function to both sides and rearranging for δ :

$$\begin{aligned} \cosh(1) &= 1 + \frac{2\delta^2}{(1-r^2)^2} \\ \delta^2 &= \frac{(\cosh(1) - 1)(1-r^2)^2}{2} \\ \delta &= (1-r^2) \sqrt{\frac{\cosh(1) - 1}{2}}. \end{aligned}$$

Given the constant $\cosh(1) \approx 1.543$, the term $\sqrt{\frac{\cosh(1)-1}{2}} \approx 0.521$. Thus, the relationship is approximately linear with respect to the term $(1-r^2)$:

$$\delta \approx 0.521 \times (1-r^2). \quad (\text{S1})$$

We now examine two specific scenarios corresponding to the AS distribution observed in our experiments:

Case 1: Core Region ($r \approx 0.4$) For nodes located near the core (typical for Tier-1 or large ISP ASes), where $r = 0.4$:

$$\delta_{\text{core}} \approx 0.521 \times (1 - 0.4^2) = 0.521 \times 0.84 \approx 0.438.$$

This indicates that in the core, a Euclidean separation of ≈ 0.44 is required to distinguish two nodes with a semantic difference of 1.

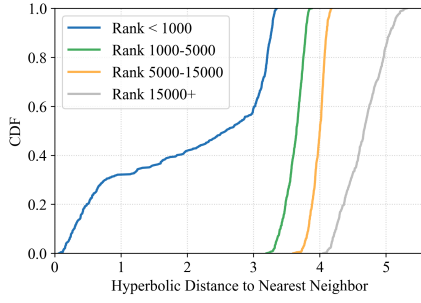


Figure S6: The CDF of the hyperbolic distance from ASes to their nearest neighbor, grouped by AS Rank. Lower-tier ASes exhibit larger semantic separation.

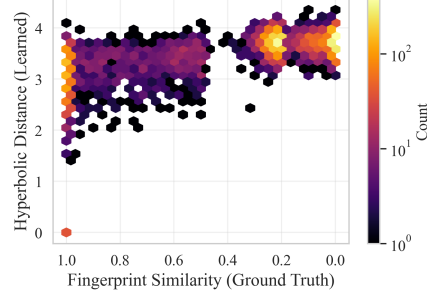


Figure S7: Correlation between propagation fingerprint similarity and hyperbolic distance. The plot reveals a distributional bifurcation around a similarity of 0.5.

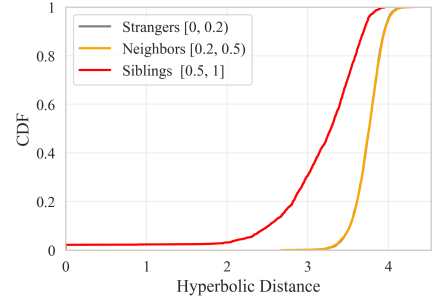
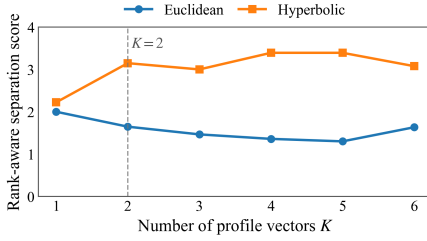
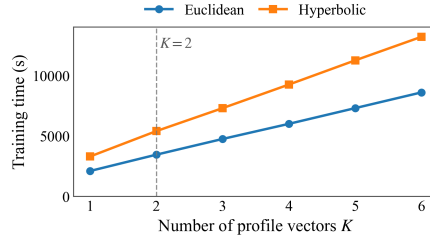


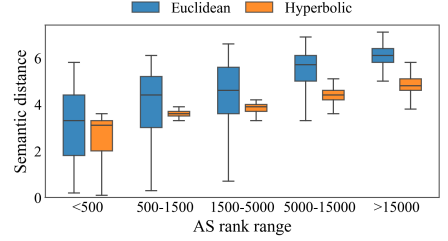
Figure S8: CDF of hyperbolic distances between profile pairs categorized by fingerprint similarity. The siblings group shows a significant leftward shift.



(a) Rank-aware separation score.



(b) Training time.



(c) Nearest-AS distances at $K = 2$.

Figure S9: Ablation of geometry and multi-vector capacity. (a) The rank-aware separation score measures how clearly nearest-AS semantic distances are stratified across AS rank groups while remaining compact within each group. (b) Training time increases steadily with the number of profile vectors K . (c) With $K = 2$, hyperbolic embeddings produce more compact and rank-consistent nearest-AS distance distributions than Euclidean embeddings.

Case 2: Periphery Region ($r \approx 0.95$) For nodes near the boundary (typical for stub ASes), where $r = 0.95$:

$$\delta_{\text{periphery}} \approx 0.521 \times (1 - 0.95^2) = 0.521 \times 0.0975 \approx 0.051.$$

Conclusion and Empirical Verification. Comparing the two cases, we observe that $\delta_{\text{core}}/\delta_{\text{periphery}} \approx 8.6$. This implies that the Euclidean space required to maintain the same semantic distinguishability shrinks by nearly an order of magnitude at the periphery of the Poincaré ball.

This capacity is empirically verified in Figure S6, which plots the CDF of the hyperbolic distance to the nearest neighbor for ASes of different ranks. As we move to stub ASes, the curves shift significantly to the right. This confirms that despite the visual compression in Euclidean space, the exponential expansion of the hyperbolic space ensures that stub ASes are not overcrowded. Instead, they enjoy sufficient semantic separation to preserve their distinguishability.

S5 Ablation and Sensitivity Analysis

We further examine how the learned semantic space is affected by the embedding geometry, the number of profile vectors K , and the embedding dimension. The geometry and vector-capacity study evaluates whether hyperbolic geometry and multi-vector profile

representations improve the structural organization of the embedding space. The dimension sensitivity study further explains the choice of the default embedding dimension.

Rank-aware separation score. We quantify the structural organization of the embedding space using a rank-aware separation score based on nearest-AS semantic distances. Let $\mathcal{R} = \{R_1, \dots, R_5\}$ denote the ordered AS rank groups from high-rank to low-rank ASes. For each rank group R_i , let m_i be the median nearest-AS semantic distance and q_i be its interquartile range. We define the rank-aware separation score as the average adjacent rank separation normalized by the average within-group dispersion:

$$\text{RSS} = \frac{\frac{1}{B-1} \sum_{i=1}^{B-1} (m_{i+1} - m_i)}{\frac{1}{B} \sum_{i=1}^B q_i}.$$

A higher score indicates that the embedding more clearly stratifies nearest-AS semantic distances across AS rank groups while maintaining compact distance distributions within each group.

Geometry and vector capacity. As shown in Figure S9a, hyperbolic embeddings provide stronger rank-aware organization than Euclidean embeddings. Increasing K from 1 to 2 substantially improves the rank-aware separation score, indicating that a single-vector profile representation is insufficient for multi-modal propagation semantics. Larger K values may further improve structural

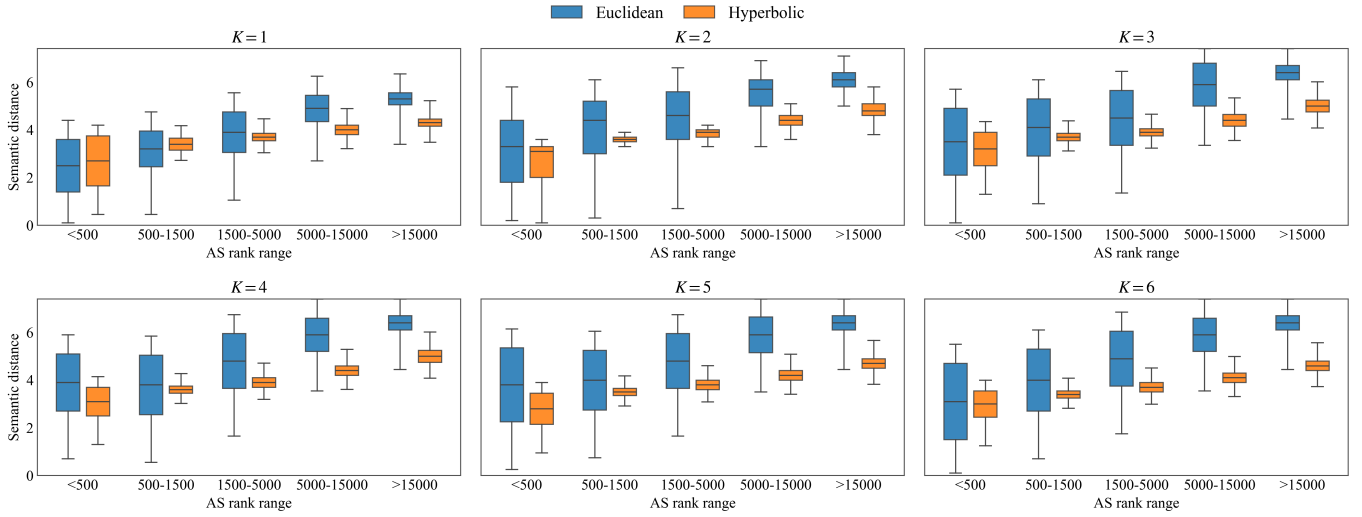


Figure S10: The complete nearest-AS distance sweep for Euclidean and hyperbolic embeddings under $K \in \{1, \dots, 6\}$.

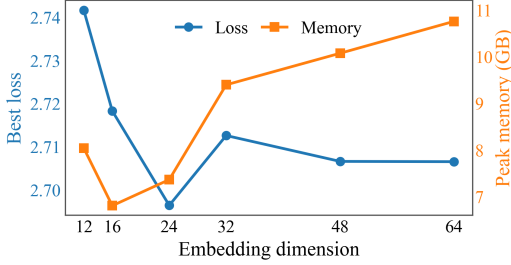


Figure S11: Sensitivity to embedding dimension under the default $K = 2$ setting. Circle markers report the best training loss using the left axis, while square markers report peak reserved CUDA memory using the right axis.

separation, but they also incur additional training cost, as shown in Figure S9b. Figure S9c further shows that, under $K = 2$, hyperbolic embeddings yield more compact and rank-consistent nearest-AS distance distributions than Euclidean embeddings.

Figure S10 provides the complete nearest-AS distance sweep for Euclidean and hyperbolic embeddings under $K \in \{1, \dots, 6\}$. Hyperbolic embeddings consistently yield more compact distance distributions within each AS rank range, indicating more stable local semantic proximity. In addition, the rank-wise medians increase from high-rank to low-rank AS groups, suggesting that the learned space preserves the hierarchical organization of AS roles.

Sensitivity to embedding dimension. We also examine the effect of embedding dimension under the default setting $K = 2$. As shown in Figure S11, increasing the dimension from 12 to 24 reduces the best training loss, suggesting that very low-dimensional embeddings have insufficient capacity. However, further increasing the dimension to 64 does not further reduce the loss and instead requires higher peak reserved memory. Training time follows a similar cost trend, increasing from 2.4k seconds at 12D to 4.6k seconds at 48D. We therefore use 24 dimensions as the default

setting, which provides a practical trade-off between optimization quality and resource cost.

Overall, moving from $K = 1$ to $K = 2$ provides sufficient capacity to capture polysemous propagation semantics, and increasing K further does not fundamentally change the rank-wise distributional pattern. Together with the dimension sweep, these results support our default configuration: hyperbolic geometry, $K = 2$, and 24-dimensional embeddings.

S6 Evaluation Setup and Parameter Selection

This section details the collector-coverage and mixed-set protocols before describing the dynamic thresholds Θ and the sensitivity calibration factor $\gamma(e)$.

S6.1 Collector Construction

We inspect the January 1, 2026 00:00 RIB and retain all collectors with nonempty observations, yielding 74 collectors: 23 from RIPE RIS and 51 from RouteViews. Coverage denotes the fraction of collectors, not the fraction of route records. For example, the nominal 20% subset contains 20.3% of collectors but approximately 31.8% of deduplicated path records because RIB sizes differ.

The subsets are deterministic, source-balanced, region-balanced, and nested. We assign RIPE RIS and RouteViews collectors separately so that no coverage level is dominated by one source. Within each source, collectors are sorted by decreasing compressed RIB size and assigned to folds by first minimizing the number of collectors from the same coarse region, then minimizing cumulative RIB bytes, and finally using the fixed fold index as a tie breaker. No random seed or anomaly-detection result enters this construction. Coverage accumulates folds, so $C_{20} \subset C_{40} \subset C_{60} \subset C_{80} \subset C_{100}$.

For each level, we reconstruct propagation graphs, fingerprints, profiles, embeddings, and historical thresholds using only that collector subset. The fixed anomaly set and the same five matched-normal sample sets are then evaluated at every level, isolating the effect of gradually adding collectors.

Table S2: Deterministic nested collector subsets.

Nominal	Collectors	RIS	RouteViews	Actual
20%	15	5	10	20.3%
40%	30	10	20	40.5%
60%	44	14	30	59.5%
80%	59	19	40	79.7%
100%	74	23	51	100%

S6.2 Mixed Evaluation Set

The aggregate coverage and threshold experiments use a fixed set of 6,000 simulated anomalies: 2,000 OH, 2,000 PM, and 2,000 RL samples. Each anomaly is matched with four legitimate routes from the January 1, 2026 RIB using the same source/transit detection branch, propagation depth, AS-rank tier, and IP version, giving 24,000 normals and a 4:1 normal-to-anomalous composition. We keep all anomalies fixed and resample only the matched normal routes five times. Metrics are computed separately for each matched-normal resampling and then averaged; repetition variability therefore means variation induced by matched-normal resampling, not different anomaly generation or model-training runs. The PM tier-depth heatmap instead uses a separate PM-only diagnostic set that is excluded from these aggregate metrics.

S6.3 Profile Consistency

Let $\mathcal{P}_c$ and $\mathcal{P}_{100}$ be the prefixes observed at coverage c and full coverage. Assignment ARI compares the two profile-membership labelings on the common-prefix set $\mathcal{P}_c \cap \mathcal{P}_{100}$; profile identifiers need not be aligned because ARI compares the induced partitions. For each common prefix p , let $A_c(p)$ be the AS set of its assigned profile at coverage c . We report the mean

$$J_{AS}(c) = \frac{1}{|\mathcal{P}_c \cap \mathcal{P}_{100}|} \sum_{p \in \mathcal{P}_c \cap \mathcal{P}_{100}} \frac{|A_c(p) \cap A_{100}(p)|}{|A_c(p) \cup A_{100}(p)|}.$$

Both consistency measures equal one at 100% coverage by reference definition. Because consistency is evaluated only on common prefixes, we also define the prefix-retention ratio

$$R_{\text{prefix}}(c) = \frac{|\mathcal{P}_c \cap \mathcal{P}_{100}|}{|\mathcal{P}_{100}|}.$$

S6.4 Detection Metrics

All coverage results pool OH, PM, and RL. Recall at 1% FPR is obtained separately for each matched-normal resampling by sweeping the decision threshold and taking the greatest recall whose empirical FPR does not exceed 1%; the five recalls are then averaged. MCC, recall, precision, and FPR in the fixed-threshold panels use $\lambda_\Theta = 1.0$. The same samples are reused across coverage levels, and metrics are first computed per resampling and then averaged. Precision is therefore precision on the controlled 4:1 mixed evaluation set, not an estimate of alert correctness under the class prevalence of a live deployment.

S6.5 Dynamic Threshold Calculation

The selection of thresholds $\Theta = \{\theta_1, \theta_2, \theta_3\}$ balances sensitivity and specificity. We derive these thresholds from the distance distributions of positive samples (historical topology) and random negative samples.

For a metric D , let τ_{err} be the threshold minimizing the sum of Type I and II errors, τ_{pos} be the 99.95th percentile of the positive distribution, and τ_{neg} be the 1.6th percentile of the negative distribution. The threshold is computed as:

$$\theta = \min(\max(\tau_{\text{err}}, \tau_{\text{pos}}), \tau_{\text{neg}}).$$

As shown in Figure S5, a profile accepts up to $\sim 1,200$ ASes at the 99.95th percentile of the positive distribution. This suggests that up to 1.6% of ostensibly unrelated ASes could actually be valid. Therefore, the upper bound τ_{neg} is introduced to ensure the model remains tolerant of unobserved legitimate routes, accounting for latent positive samples missing from the training data.

S6.6 Radial Sensitivity Calibration

As discussed in §4.4 of the paper, the Poincaré ball space can make highly central ASes appear close to many profiles, which may reduce the sensitivity of self-origination checks. We therefore introduce a radial calibration factor $\gamma(\mathbf{e})$ that tightens the decision boundary for central ASes according to their hyperbolic norm.

We identify two centrality breakpoints from the empirical AS-norm distribution. Since smaller hyperbolic norms indicate more central ASes, we sort ASes in ascending order of $\|\mathbf{e}\|$ and set $r_{\text{core}} = 0.8$ and $r_{\text{edge}} = 0.95$. These two cutoffs approximately correspond to the top 0.5% and top 10% most central ASes, respectively, which are roughly the top 500 and top 10,000 ASes in our embedding.

Based on these empirical cutoffs, we define $\gamma(\mathbf{e})$ as:

$$\gamma(\mathbf{e}) = \begin{cases} 0.8, & \|\mathbf{e}\| < r_{\text{core}}, \\ 0.8 + \frac{1.0-0.8}{r_{\text{edge}}-r_{\text{core}}} (\|\mathbf{e}\| - r_{\text{core}}), & r_{\text{core}} \leq \|\mathbf{e}\| \leq r_{\text{edge}}, \\ 1.0, & \|\mathbf{e}\| > r_{\text{edge}}. \end{cases}$$

The calibration is strongest for ASes in the extreme core region, gradually weakens within the broader core region, and becomes inactive outside this region. The floor factor of 0.8 is empirically derived from the ratio of average propagation distances between Tier-1 and stub ASes, ensuring that central ASes are not systematically under-detected due to hyperbolic centrality bias.

S6.7 Operational Threshold Sensitivity

The thresholds above are derived from the geometry of historical positive and random negative pairs. Public BGP observations are incomplete, so some nominal negative or normal samples may be unobserved legitimate relationships; the sweep therefore focuses on the operating-point trade-off among recall, precision, and alert volume. We evaluate a multiplicative threshold scale λ_Θ on five matched-normal resamplings over the same 6,000 anomalies at 100% collector coverage. For the resampling generated with seed 42, five-fold validation maximizes MCC near $\lambda_\Theta = 0.80$ and the one-standard-error rule selects the more conservative 0.85; the remaining matched-normal resamplings verify the trend.

Smaller scales improve recall and increase operator load. Larger scales reduce FPR and alert volume together with recall. We retain

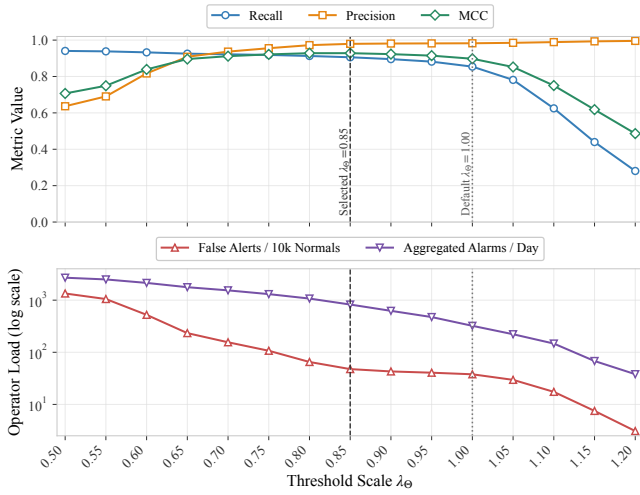


Figure S12: Sensitivity to the threshold scale λ_Θ . The upper panel reports candidate-level recall, precision, and MCC on the mixed evaluation set. The lower panel reports false alerts per 10,000 normal samples and aggregated alarms per day on a logarithmic scale. Vertical lines mark the validation-selected point (0.85) and deployed default (1.0).

$\lambda_\Theta = 1.0$ as the deployed operating point because it follows the distribution-derived thresholds and produces a conservative alert volume. The label-based sweep identifies 0.85 as the statistical balance for this evaluation set.

ASPA validation. On the same 2,000 simulated leaks, recall is 0.678 overall, 0.701 for 385 ASPA-covered samples, 0.710 for 376 ASPA-Invalid paths, and 0.671 for 1,624 ASPA-Unknown paths. ASPA coverage means that at least one relevant provider attestation is available, whereas Invalid and Unknown are route-validation outcomes. Thus, Invalid is a subset of covered; the Unknown group contains 1,615 uncovered paths and nine covered but inconclusive paths, and is not the complement of the covered group. The at-most 3.2-point gap is consistent with the experiment design: CAIDA/ASPA affect test labels or validation, while the encoder learns exclusively from BGP observations.

S6.8 Preprocessing and Training Scaling

We measure the complete input-preparation and training workflow on the same deterministic nested collector subsets used in the coverage experiment. From 20% to 100% input, the number of unique paths increases from 139.1 million to 436.8 million, and the number of training triplets increases from 45.9 million to 169.9 million.

Figure S13 shows that parser and dataset-building time grows monotonically from approximately 10 to 41 minutes. In contrast, fixed-work training through epoch 100 remains within approximately 9–12 minutes. Early-stop training and end-to-end time are non-monotonic because different input scales converge at different epochs; in particular, the 80% input reaches its stopping criterion later than the 100% input. Cached parser outputs, datasets, and model checkpoints are disabled, and each scale represents one complete measured run. At 100% input, the complete preprocessing

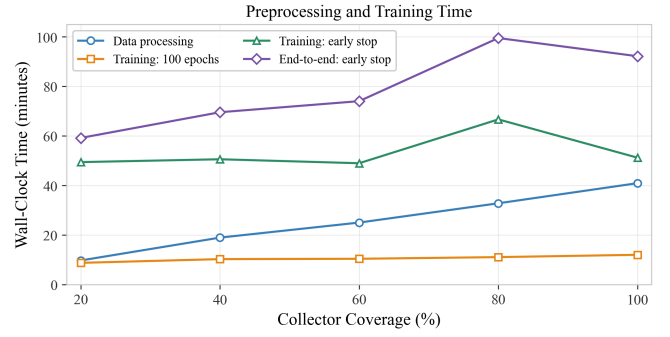


Figure S13: Preprocessing and training scaling over nested collector inputs. The figure reports wall-clock time for the parser plus dataset/anchor construction, training through epoch 100, training through the patience-based early stop, and the end-to-end preprocessing time used in Table 2 of the main paper, i.e., data processing plus early-stop training.

workflow—data processing plus patience-based training—takes 1.53 hours, and remains below two hours at every measured scale. These results therefore support an empirical input-scaling comparison, not a linear or asymptotic complexity claim.

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